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Code to Connect
Hackathon 2026

AI RISK ENGINE

From Financial News to Portfolio Decisions

Turning unstructured financial information into explainable, systematic portfolio risk signals.

Candidate	**Mridul Rajgaria**
Institution	Vellore Institute of Technology, Bhopal
Event	Code to Connect Hackathon 2026 • Final Round
Implemented application	Module A — Tactical Index Rebalancer
Portfolio universe	14 large-cap equities

NEWS HEADLINE • ILLUSTRATIVE EXAMPLE

Acme Corp faces regulatory probe over accounting practices

↓

STRUCTURED RISK SIGNAL

event_type	**Regulatory**
sentiment_score	-0.80
impact_score	**7.85 / 10**
trading_action	REDUCE

Illustrative values, not an application output

Text in, structured signal out, portfolio weight adjusted.

Financial information arrives as text. A portfolio needs numbers.

RAW INFORMATION • ILLUSTRATIVE EXAMPLE

News headline

Acme Corp faces regulatory probe over accounting practices

Social post

$ACME this probe is going to get ugly. Not touching it.

Market data

ACME daily prices (yfinance)

STRUCTURED RISK SIGNAL • ILLUSTRATIVE VALUES

`event_type`	**Regulatory**	1 of 8 classes, rule-based
`sentiment_score`	`-0.80`	FinBERT tone, range [-1.0, +1.0]
`confidence`	`0.90`	softmax conviction, [0.0, 1.0]
`impact_score`	`7.85`	severity, [1.0, 10.0]: Event 35% + Sentiment 30% + Credibility 20% + Volume 15%
`trading_action`	REDUCE	downstream suggestion: BUY, REDUCE or HOLD

The engine converts unstructured financial text into consistent, explainable information that a portfolio strategy can consume.

Information overload

Too many headlines and posts to assess consistently by hand.

Tweets, NewsAPI and GDELT are processed into one schema with cross-label deduplication.

Black-box distrust

A score is less useful if its reasoning cannot be inspected.

A linear impact score (1.0 to 10.0) built from 4 named, audit-ready components, configurable via YAML.

Execution drag and noise

Reacting to every small change can cause excessive trading.

Rebalancer with ±0.20 deadband, EMA half-life 2.0 days and a 10% daily turnover cap.

Reproducibility

A reviewer needs to reproduce the result without live APIs.

Deterministic committed sample datasets (<5 MB): full audit and replay with zero API keys or live downloads.

One pipeline: collect, understand, decide, deliver

1 Collect

Gather headlines, social posts and market data.

Kaggle Stock Tweets (2021–22), NewsAPI, GDELT 2.0, yfinance • 14 core + 7 news-only equities

2 Understand

Identify the company, read sentiment, classify the event.

Normalization, cashtag linking, FinBERT (CPU), event classifier • 1–10 impact severity

3 Decide

Turn the signal into a bounded weight adjustment.

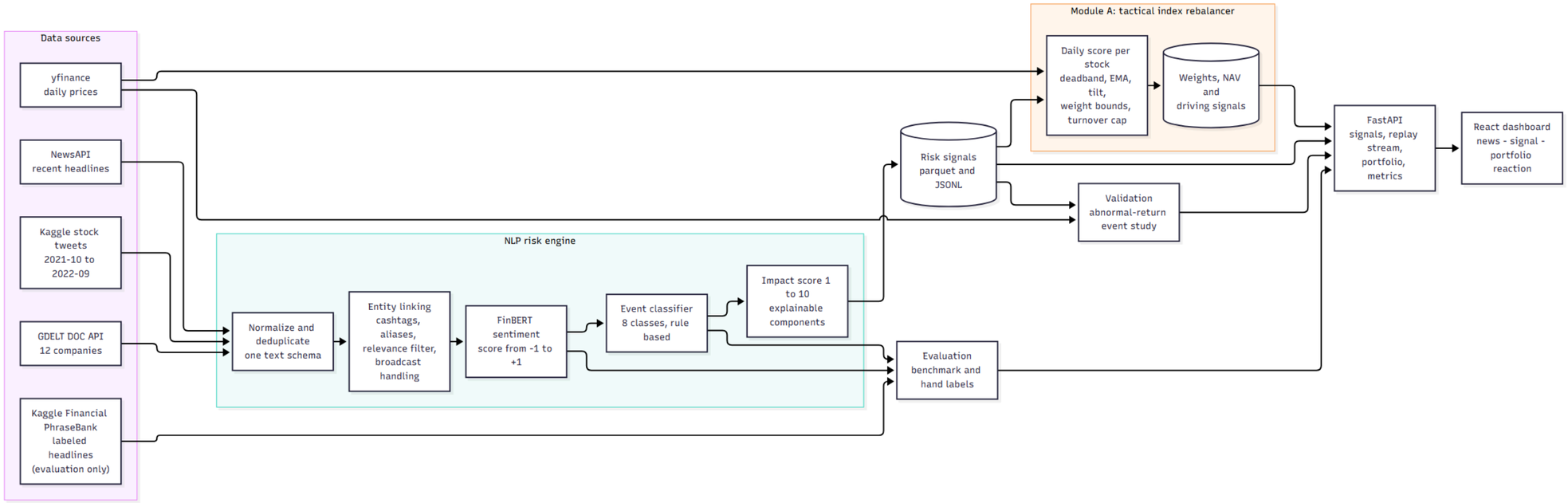
EMA smoothing, tilt k=1.5, negative multiplier 1.25x • weights [3.57%, 14.29%], 10% turnover cap

4 Deliver

Show the result and its reasoning in a dashboard.

FastAPI, Server-Sent Events, React 18, Apache ECharts • News Stream, Signal Inspector, Portfolio Reaction

Technical architecture, as implemented (unchanged from the original submission)



How one headline becomes a risk signal

Illustrative example: values follow the published formula but are not an application output.

1 • Headline

Acme Corp faces regulatory probe over accounting practices

First cleaned: template spam stripped, URLs normalised, HTML entities decoded.

2 • Entity linking

Acme Corp → $ACME

The system determines which company the text refers to.

cashtags, aliases, name dictionaries, broadcast splitting

3 • Sentiment

`-0.80`



The model estimates whether the language is positive or negative. Confidence 0.90.

FinBERT (ProsusAI/finbert), SQLite cache

4 • Event type

Regulatory

Classifies the event: earnings, regulatory action, credit event, etc.

8 rule-based classes • precision 1.0 on Credit / Regulatory / Earnings triggers

5 • Impact scoring

Event severity	`35% Regulatory = 0.9`
Sentiment strength	`30% |s| × c = 0.72`
Source credibility	`20% GDELT 0.85`
Mention volume	`15% 0.40 (illustrative)`

Credibility: GDELT 0.85, NewsAPI 0.80, Kaggle Tweets 0.50. Severity: Regulatory/Credit 0.9, Earnings 0.7, Product 0.5, Other 0.3. Volume: log-scaled tweet density vs rolling median.

IMPACT-SCORE FORMULA

$$\text{Impact} = 1.0 + 9.0 \times \sum(w_i \times \text{Component}_i) \in [1.0, 10.0]$$

$$1.0 + 9.0 \times (0.35{\times}0.90 + 0.30{\times}0.72 + 0.20{\times}0.85 + 0.15{\times}0.40) = 1.0 + 9.0 \times 0.761 = \mathbf{7.85}$$

Combines event importance, sentiment, source reliability and mention activity into one consistent severity measure.

What the score means, and does not

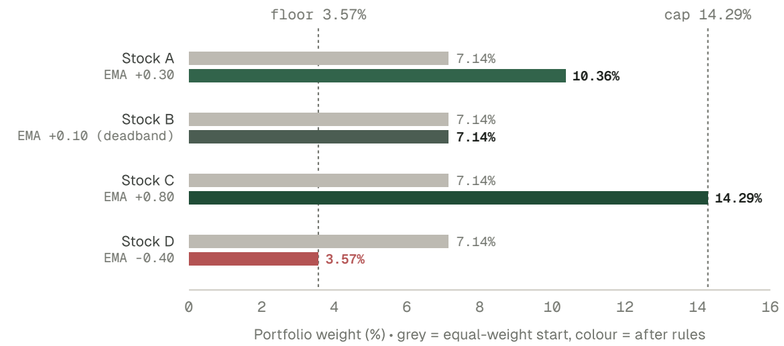
It ranks how severe a piece of text looks. It is not a probability of a price move or a promise of trading return.

Every score on the dashboard can be unpacked via “Show the math” into its 4 raw component values.

How a risk score nudges portfolio weights, within limits

If all 14 stocks start with equal weights, each is 1 / 14 ≈ 7.14% of the portfolio. Signals tilt that baseline; rules keep the tilt bounded.

Equal-weight portfolio → Risk signal → Bounded adjustment



Portfolio weight (%) • grey = equal-weight start, colour = after rules

> Conceptual example, not an application output. Four hypothetical stocks (A–D). Bars apply the raw formula, deadband and weight bounds only (Stock C is clamped from 15.71% to the 14.29% cap); no renormalisation, turnover cap or negative multiplier. Not an optimal-allocation claim.

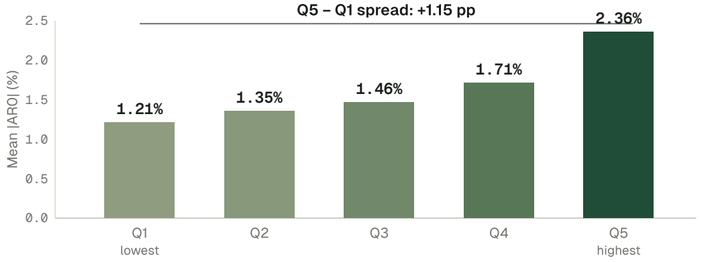
$$w_{raw,i} = w_{base} \times (1.0 + k \times EMA_i)$$

$w_{base} = 7.14\% \cdot k = 1.50$

EMA smoothing half-life 2.0 days	A brief burst of news cannot cause an unstable, immediate change.
Deadband ±0.20	Small signals, likely mostly noise, are zeroed out.
Negative multiplier 1.25×	Negative headlines get a 25% higher penalty (asymmetric risk aversion).
Weight bounds 3.57% (0.5×) to 14.29% (2.0×)	No stock becomes too small or too large (e.g. concentration in TSLA).
Turnover cap 10% / day	Limits how much the portfolio can trade in one day.
Transaction costs 5 bps	Trading is not free: cost is deducted from net NAV.
Fallback equal weight	Missing market data reverts weights to 7.14%.

Higher impact scores went with larger event-day price moves

Event study across 2,072 ticker-day observations (2021–2022), with block bootstrap confidence intervals.



Impact-score quintile (N = 2,072; about 414 per bucket)

	Q1	Q2	Q3	Q4	Q5	Q5 – Q1
Count	415	414	415	414	414	2,072
Mean impact	2.70	3.50	4.63	5.15	6.36	+3.66
Mean |CAR(0,2)|	2.46%	2.52%	2.81%	3.10%	**3.89%**	**+1.43%**

What was compared

We grouped observations into five buckets, lowest to highest impact score, and compared the average absolute abnormal price move (|AR0|: the event-day move beyond what the market explains) in each.

What appeared

Q1 averaged 1.21%, Q5 2.36%. Higher impact-score groups had larger average absolute event-day abnormal moves in this sample.

How strong

ρ = 0.159	Spearman rank correlation, p = 3.08e-13: a positive but modest monotonic association.
ρ = 0.116	Partial correlation (p = 1.05e-07), controlling for recent 20-day historical volatility.
p = 0.0010	Placebo test: 1,000 within-ticker shuffles of the scores.

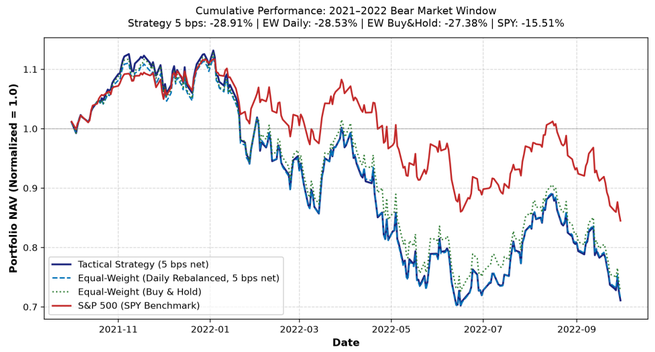
What it does not establish

Correlation is not causation. The score does not predict each price move, and |AR0| measures size, not direction.

The strategy lost money and trailed every benchmark

252 trading days, Oct 2021 to Sep 2022, during the tech-sector downturn. 5 bps net transaction friction.

Cumulative performance, normalised NAV (start = 1.0), full axis range shown



The strategy was tested during a difficult historical period. It did not outperform the reported benchmarks in this backtest. Its result sat near the middle of the placebo distribution (48.5th percentile), so this experiment did not establish a robust advantage over randomised alternatives.

Strategy	Return	Sharpe	Max DD
Tactical (5 bps)	-28.91%	-0.9209	-37.95%
Equal-wt daily	-28.53%	-0.9092	-37.39%
Equal-wt buy & hold	-27.38%	-0.9323	-35.56%
SPY (S&P 500)	-15.51%	-0.7085	-24.37%

Sharpe: return per unit of volatility (risk-free rate = 0). Max DD: largest peak-to-trough fall.

Honesty diagnostics

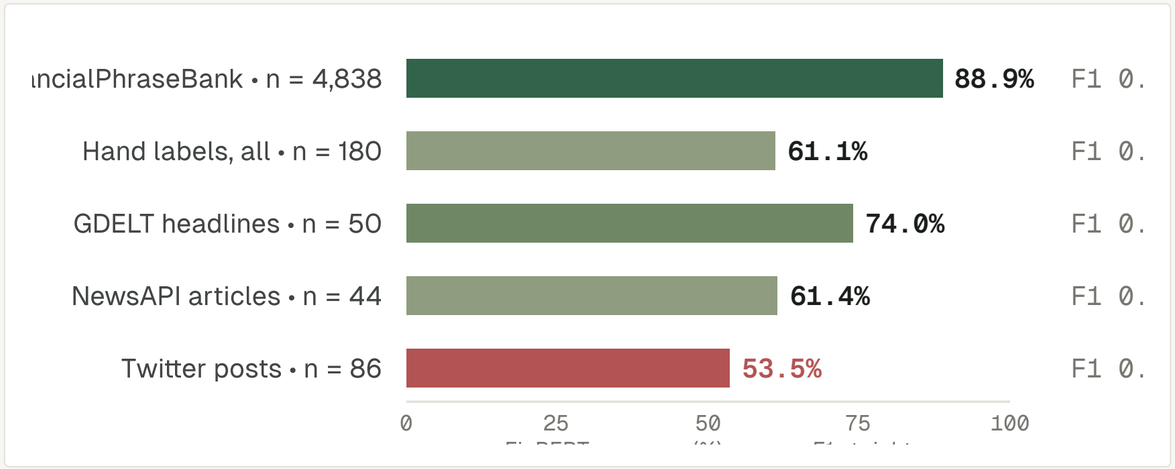
Placebo
200 runs
48.5th percentile: near the centre of the random distribution.

TSLA exclusion
-28.65% net. Dropping the stock with ~70% of tweet volume does not change the result.

Sensitivity grid
9 cells
-28.56% to -29.20% across tilt $k \in [0.75, 3.0]$ and half-life [1, 5] days: flat.

Where the model is weak, and what could come next

MODEL LIMITATION • MEASURED RESULT



A model that performs well on curated financial headlines may perform differently on informal social posts and other real-world text. Informal language, ambiguity and writing style are plausible causes, but they are hypotheses, not tested here.

FinBERT is precise on bad news (0.85), useful for risk detection, but misses subtle positive analyst targets (positive recall 0.30). Accuracy = share of texts labelled correctly; macro-F1 averages per-class scores.

PROPOSED EXTENSION • MODULE B, WHOLESALE LOAN STRESS TESTING

Proposed, not implemented or validated.

Risk signal • the same RiskSignal feed, zero engine modification

↓ **Credit-risk assessment** • severe Credit Event / Regulatory triggers would raise PD; macro severities would lower sector recovery (raise LGD)

↓ **Scenario analysis** • re-estimate exposures after sudden news shocks

↓ **Potential stress-testing application** • capital adequacy and regulatory stress reporting

$$EL = EAD \times PD_{adjusted} \times LGD_{adjusted}$$

Expected loss = how much is at risk × how likely default is × how much is lost if it happens.

PD Probability of Default • **LGD** Loss Given Default • **EAD** Exposure at Default

The project demonstrates an end-to-end path from financial text to explainable portfolio signals. Further validation is needed before treating those signals as a reliable source of investment advantage.